UNIVERSITY OF SZEGED

Faculty of Science and Informatics

Department of Computer Science

Project Title:

Machine Learning Based Car Price Prediction

Course: Machine Learning (Practice) (2025/2026 Spring Semester)

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Abstract / Project Overview:

This report presents the first milestone of the "Machine Learning Based Car Price Prediction" project. The primary goal is to explore, clean, and analyze a comprehensive automotive dataset to understand the key factors influencing vehicle market values. By identifying data quality issues such as missing values and outliers, and performing exploratory data analysis (EDA), this phase establishes the necessary foundation for building robust machine learning regression models in the subsequent stages of the project.

1. TASK EXPLORATION AND PROBLEM IDENTIFICATION

1.1.Description of the Task

The objective of the **"Machine Learning Based Car Price Prediction"** project is to develop a computational model that estimates the market value of a used vehicle. Car valuation is a complex process influenced by a multitude of factors, and this project aims to automate that process using historical sales data.

The core challenge is to identify patterns within a dataset of roughly 19,000 car listings. The model must learn how different attributes (such as how old the car is or how much it has been driven) contribute to its final price.

1.2. Identification of the Task Type

In Machine Learning terms, this project is classified as a **Supervised Learning** task, specifically a **Regression** problem.

- **Why Supervised Learning?** Because we train the model using a dataset that includes both the input features (car specifications) and the correct output labels (actual prices).
- **Why Regression?** Because the target variable, Price, is a **continuous numerical value**. The model does not predict fixed categories (like "Cheap" or "Expensive") but rather calculates a specific numeric amount (e.g., $14,750). The mathematical goal is to minimize the error between the predicted price and the real market price.

1.3. Background of the Dataset

I. **Source:** The dataset, titled "Car Price Prediction Challenge," is sourced from the Kaggle platform.

II. **Origin:** The data contains real-world listings scraped from online automotive marketplaces.

III. **Motivation:** Automating car valuation has significant practical benefits:

   a) **For Consumers:** Helps ensure a fair price during buying or selling.

   b) **For Businesses:** Enables dealerships to instantly value trade-ins and optimize their inventory.

   c) **As a Case Study:** Provides a realistic scenario for handling messy, real-world data in a computer science context.

2. BACKGROUND AND MOTIVATION

2.1. Dataset Provenance (Where does it come from?)

The dataset utilized in this project, titled **"Car Price Prediction Challenge,"** is a collection of automotive sales records sourced from the **Kaggle** platform. It was originally made available by the user **DeepContractor** to provide a realistic scenario for regression-based machine learning tasks.

- **Source Characteristics:** The dataset contains **19,237 unique samples**, reflecting a broad snapshot of the secondary car market.
- **Data Nature:** Unlike perfectly cleaned academic datasets, this data is "raw" and "noisy." It represents real-world listings scraped from digital marketplaces. This means it includes human-entered errors, inconsistent formatting (e.g., "km" strings inside numeric columns), and missing values.
- **Scope:** It covers various manufacturers (Toyota, BMW, Mercedes-Benz, etc.), multiple fuel types, and various mechanical specifications, making it a diverse representation of the automotive industry.

2.2. Motivation Behind Solving the Task

The motivation for developing a **Machine Learning Based Car Price Prediction** model is rooted in both economic necessity and technological application.

2.2.1. Economic Transparency and Fair Valuation

In the used car market, there is a significant **information asymmetry**. Sellers often have more information about the vehicle's true condition than buyers.

- **For Buyers:** An automated ML model provides an objective "fair price" baseline, preventing them from overpaying based on seller intuition.
- **For Sellers:** It helps in setting competitive prices that are grounded in actual market data, increasing the likelihood of a quick sale.

2.2.2. Business Process Automation

- **Dealerships:** Professional car dealers handle hundreds of trade-ins. Manually appraising each car is time-consuming. An ML model provides an instant valuation, speeding up the business cycle.
- **Financial Services:** Insurance companies and banks require precise vehicle valuations to calculate insurance premiums or car loan limits. An automated system reduces the overhead costs of manual appraisal.

2.2.3. Data Science Challenge (Technical Motivation)

From a Computer Science perspective, the motivation is to master the **End-to-End Machine Learning pipeline**. This specific dataset is chosen because it requires:

- **Complex Data Cleaning:** Converting strings to integers and handling missing values.
- **Feature Engineering:** Deriving insights like "Car Age" from the production year.
- **Outlier Management:** Identifying and removing unrealistic prices (e.g., $1 listings) that would otherwise confuse the model.

3. DATASET DESCRIPTION

3.1. Target Variable

The primary objective of the **"Machine Learning Based Car Price Prediction"** model is to predict a specific value:

- **Target:** Price
- **Type:** Numerical (Continuous)
- **Description:** This column represents the final selling or listing price of the vehicle. In the raw dataset, prices vary significantly, ranging from very low values (potential outliers) to luxury vehicle prices. Predicting this value accurately is the core "Regression" task of the project.

3.2. Input Features

The dataset contains various attributes that influence the final price. These are categorized into Categorical (text-based) and Numerical (number-based) features.

Feature Name	Data Type	Description
Levy	Numerical (Object*)	Tax or financial obligation on the vehicle.
Manufacturer	Categorical	The brand of the car (e.g., Toyota, BMW, Ford).
Model	Categorical	The specific model name (e.g., Camry, X5, Focus).
Prod. year	Numerical	The year the vehicle was manufactured.
Category	Categorical	The body style (e.g., Sedan, Jeep, Hatchback).
Leather interior	Boolean	Indicates if the interior is leather (Yes/No).
Fuel type	Categorical	Type of fuel used (Petrol, Diesel, Hybrid, etc.).
Engine volume	Numerical (Object*)	Displacement of the engine (e.g., 2.0, 3.5 Turbo).

Feature Name	Data Type	Description
Mileage	Numerical (Object*)	Total distance driven (recorded in km).
Cylinders	Numerical	Number of engine cylinders.
Gear box type	Categorical	Type of transmission (Automatic, Manual, Tiptronic).
Drive wheels	Categorical	Front, Rear, or 4x4 (All-wheel drive).
Wheel	Categorical	Steering wheel position (Left wheel or Right wheel).
Color	Categorical	The exterior color of the vehicle.
Airbags	Numerical	Total number of airbags in the car.

```python
import pandas as pd

# Loading the dataset
df = pd.read_csv('car_price_dataset.csv')

# Inspecting the first 5 rows and column types
print(df.head())
print(df.info())
```

Figure 1: This code snippet is used for initial data loading and structural inspection. By using **df.head()**, we see the actual values (like "km" strings and dashes), while **df.info()** reveals the data types. This step is crucial because it confirms that most numerical features are incorrectly stored as "Object" types, providing the technical justification for the data cleaning required in the next sections.

3.3. Structure of the Dataset

Understanding the raw structure is essential before applying any Machine Learning algorithms.

I. **Number of Samples:** The dataset consists of **19,237 individual records (rows)**.

II. **Number of Columns:** There are **18 columns** in total (17 features + 1 target).

III. **Data Distribution (Target):** Initial observation of the Price column shows a **right-skewed distribution**. Most vehicles are concentrated in the mid-price range, with a few high-end luxury vehicles creating a "long tail" in the data.

IV. **Train and Test Set Availability:** The original Kaggle dataset does not provide a pre-divided train and test set. Therefore, as part of this project, I will implement a **randomized split (typically 80% train / 20% test)** using the train_test_split method from the Scikit-Learn library to ensure unbiased model evaluation.

4. DATA QUALITY ASSESSMENT

Before proceeding with model training, a rigorous evaluation of the dataset's integrity is mandatory. Real-world data often contains noise and inconsistencies that can lead to biased predictions. This section outlines the primary quality issues identified in the dataset.

4.1. Identification of Hidden Missing Values

A standard programmatic check (e.g., isnull().sum()) suggests that the dataset is complete. However, a manual inspection reveals **"Hidden Missing Values"** where data is replaced by placeholder characters.

- **The Levy Feature:** Approximately **30.2%** of the entries in the Levy column contain a dash (-) instead of a numeric tax value. Since these are strings, they are not automatically detected as null values by the system.
- **Placeholder Prices:** There are several records where the Price is listed as **$1**. These are functionally missing values (placeholders) that do not reflect the actual market value of the vehicle and must be treated as erroneous data.

4.2. Structural Inconsistencies and Type Mismatch

Data quality is significantly impacted by the storage of numerical information in string (Object) formats. This prevents direct mathematical analysis.

1. **Mileage:** Every entry in this column includes the "km" suffix (e.g., "120000 km"). This unit marker forces the entire column to be treated as text, requiring string stripping and type casting to integers.
2. **Engine Volume:** This feature is inconsistent; some entries contain only numbers (e.g., "2.0"), while others include categorical information like "Turbo" (e.g., "2.0 Turbo"). This requires feature splitting to separate displacement from engine type.
3. **General Formatting:** Columns such as Doors are stored in ranges (e.g., "04-May") due to Excel-style formatting errors during data collection, which requires manual correction.

4.3. Quality Metrics Summary

Feature	Issue Type	Proportion	Required Action
Levy	Hidden Nulls (-)	~30%	Median Imputation
Mileage	Unit Suffix (km)	100%	String Stripping
Price	Extreme Outliers	~2%	Statistical Trimming
Engine Vol.	Mixed Text/Numeric	~10%	Parsing & Splitting

4.4. Preliminary Conclusion on Data Integrity

The dataset is "Real-World Messy," meaning it requires significant pre-processing before any machine learning algorithm can be applied. The high percentage of missing values in the Levy column implies that dropping these rows would result in a massive loss of information. Instead, an **imputation strategy** is necessary. Furthermore, correcting the data types for Mileage and Engine Volume is the highest priority for the next phase of the project.

```python
# 1. Clean Mileage: Remove 'km' and convert to integer
df['Mileage'] = df['Mileage'].str.replace(' km', '').astype(int)

# 2. Clean Levy: Replace '-' with NaN to calculate true missing proportion
df['Levy'] = df['Levy'].replace('-', np.nan).astype(float)

# 3. Clean Engine Volume: Extract only the numeric part
df['Engine volume'] = df['Engine volume'].str.split(' ').str[0].astype(float)

print(f"Missing Values after correction:\n{df.isnull().sum()}")
```

Figure 2: *The data cleaning pipeline. By stripping units and handling hidden nulls, we transform 'Object' types into 'Float/Int' types, making them compatible with Scikit-Learn algorithms.*

5. PRELIMINARY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is performed to uncover patterns, detect anomalies, and check assumptions with the help of summary statistics and graphical representations.

5.1. Basic Statistical Summary

Before cleaning, a statistical overview of the numerical features provides a baseline for our data distribution.

Statistical summary of numerical features

print(df.describe())

Key Findings:

- **Price Variance:** The Price column shows an extremely high standard deviation, suggesting the presence of luxury cars or data entry errors (outliers).
- **Average Mileage:** The mean mileage is significantly high, which is typical for a used car dataset, but the maximum values require further investigation for potential "fake" entries.

5.2. Distribution of the Target Variable (Price)

The distribution of the target variable is the most critical factor for a regression model.

- **Analysis:** The Price distribution is heavily **right-skewed**. Most vehicles are concentrated in the lower price brackets, while a few "long-tail" luxury items exist.
- **Action:** To help the model learn better, a **Log Transformation** may be considered in the next phase to normalize this distribution.

5.3. Feature Correlations (Heatmap)

To understand which features (e.g., Year, Engine Volume) have the strongest impact on the Price, we generate a **Correlation Heatmap**.

```python
import seaborn as sns
import matplotlib.pyplot as plt

# Plotting the correlation matrix
plt.figure(figsize=(10, 8))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```

(Figure 3)

- **Observation:** We expect a strong positive correlation between Prod. year and Price (newer cars are more expensive) and a negative correlation between Mileage and Price (highly driven cars are cheaper).
- **Multicollinearity Check:** The heatmap also helps us identify if two features are too similar, which could confuse the model.

5.4. Pairwise Relationships (Mileage vs. Price)

A scatter plot is used to visualize the "price decay" as mileage increases.

Insight: There is a visible trend where price drops as mileage increases. However, the "noise" (scattered points) indicates that other factors like Manufacturer and Airbags also play a significant role in determining the final value.

6. SUMMARY OF FINDINGS

This section summarizes the technical insights and data states identified during the Milestone 1 exploration phase:

- **Task Identification:** The project is strictly defined as a **Supervised Regression** task because the target variable (Price) is a continuous numerical value rather than a discrete category.
- **Data Quality Issues:** The dataset is "Real-World Messy," featuring **30.2% hidden missing values** in the Levy column (represented by dashes) and 100% type mismatches in Mileage due to unit suffixes.
- **Price Drivers:** Preliminary analysis through correlation heatmaps confirms that **Production Year** and **Mileage** are the most significant predictors of value, showing strong positive and negative correlations respectively.
- **Distribution Insights:** The Price variable is heavily **right-skewed**, indicating the presence of luxury outliers and a potential need for log-transformation in the modeling stage to normalize the target.
- **Outlier Detection:** Initial inspection revealed unrealistic entries, such as vehicles priced at **$1**, which must be removed to ensure the integrity and accuracy of the machine learning model.

Conclusion: The exploratory phase has successfully mapped the data challenges, establishing a clear pre-processing roadmap (cleaning, imputation, and scaling) for the next milestone.